



Stop writing prompts.

Design the system that writes them.

Addy Osmani's loop engineering,
mapped to AWS — 6 components

THE SHIFT

A prompt is a task.

A loop is a system.

Prompting

you type "go" every single time

Loop engineering

the system fires itself, then hands you decisions

Four moving parts, self-orchestrating:

trigger → **memory**

→ **tools** → **agent**

"A recursive objective where AI iterates until complete." — Osmani

COMPONENT 1 · AUTOMATIONS

The trigger that replaces you typing.

AWS service

EventBridge + Lambda

EventBridge wakes the loop on a cron tick or an event pattern.

Lambda is the dispatcher: it sets the objective and holds the circuit breaker.

No human types "go." That's the loop.

COMPONENT 2 · WORKTREES

Parallel work without collisions.

AWS service

CodeBuild / ECS

One container or task per agent.
Filesystem-isolated state, clean
start, torn down after.

Shared mutable state is how you
get a mess you can't debug.

Isolation, mapped. Not git worktrees, cloned.

COMPONENT 3 · SKILLS

What the agent knows how to do.

AWS service

Bedrock Knowledge Bases

Reusable project knowledge the agent loads into context, so it doesn't relearn your conventions.

Smart Parsing pre-computes chunks and embeddings. Structured S3 too.

S3 for every-run facts. KB for the big corpus.

COMPONENT 4 · MCP

**Live AWS state,
no glue code.**

AWS service

AWS MCP Server

GA · May 2026

MCP became the default way agents reach tools. It replaced the custom integrations.

The orchestrator calls out without you maintaining connectors.

IAM guards access. CloudTrail logs every call.

COMPONENT 5 · SUB-AGENTS

Specialists with a coordinator.

AWS service

Bedrock multi-agent + AgentCore

An orchestrator routes work to specialist agents. AgentCore holds it together in production.

200 TPS

per agent

5,000

sessions per account

Model the session ceiling first.

COMPONENT 6 · MEMORY

What survives between runs.

S3

long-term

context files, artifacts

DynamoDB

short-term

fast state, checkpoints

The session ID is the thread. It ties the trigger to state to context across every dispatch.

Without it, six Lambda runs are six strangers, not one loop.

Cycle two knows what cycle one did.



**The loop outlives
the session.**

That's the point.

Full component-by-component map →
Featured section on my profile.

**Which component is your
weakest link today?**

Build the loop. Stay the engineer.